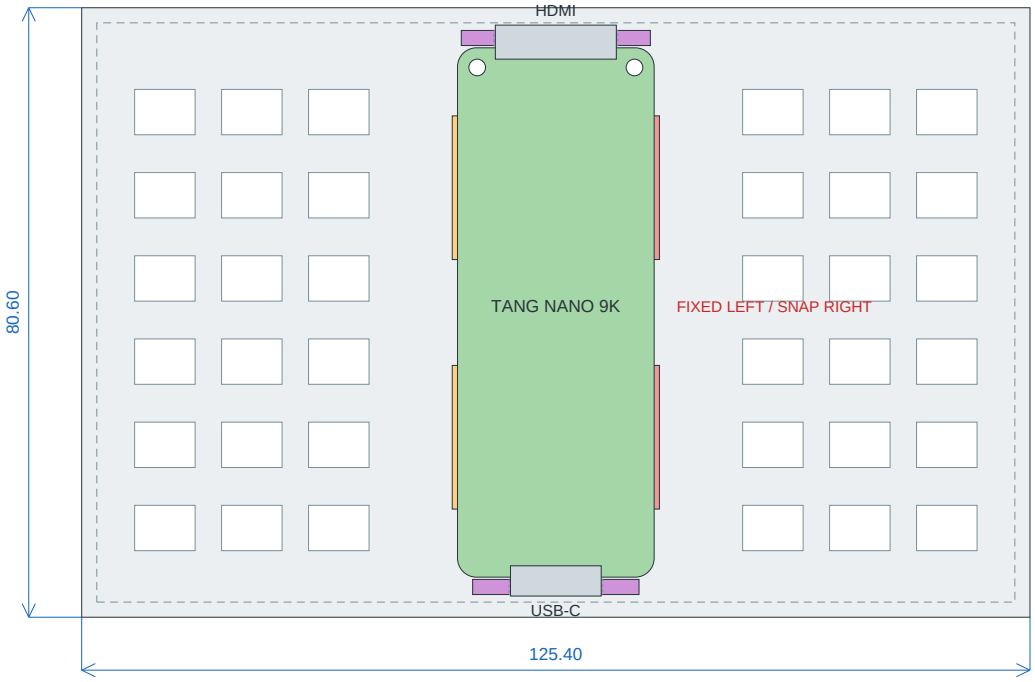
Tang Nano 9K + 5-inch LCD panel case

Assembly placement — rear-facing views at true 1:1 scale

SHEET 1/10

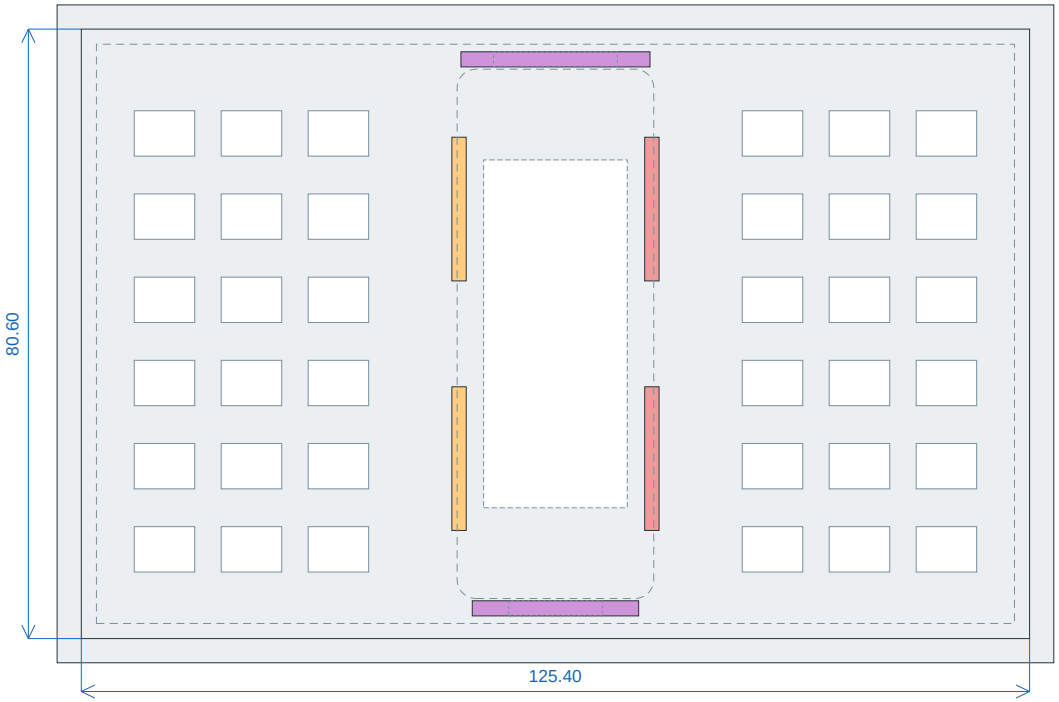
UNITS mm SCALE 1:1

STEP 1 INSTALL PCB IN REAR-COVER RAILS



Component side faces LCD; microSD side faces rear service aperture.

STEP 2 FLIP COVER AND SNAP INTO CHASSIS



Dashed: Tang Nano 9K behind rear cover

HDMI = TOP / USB-C = BOTTOM / microSD = REAR APERTURE

CALIBRATION 100.0 mm

PRINT: A4 LANDSCAPE / ACTUAL SIZE (100%) / FIT TO PAGE OFF. VERIFY THE 100 mm LINE.

Printed part A: front chassis

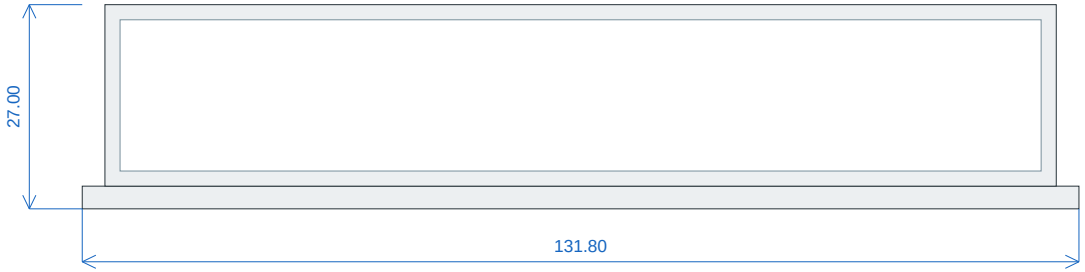
Front outline, centre section, and panel cutout — true 1:1

SHEET 2/10

UNITS mm SCALE 1:1

PANEL CUTOUT 126.40 x 81.60

CENTRE SECTION X-Z / CHASSIS DEPTH 27.00



STL variants: 1.5 / 2.0 / 3.0 mm panels. Outer geometry is common; snap catch depth changes.

CALIBRATION 100.0 mm

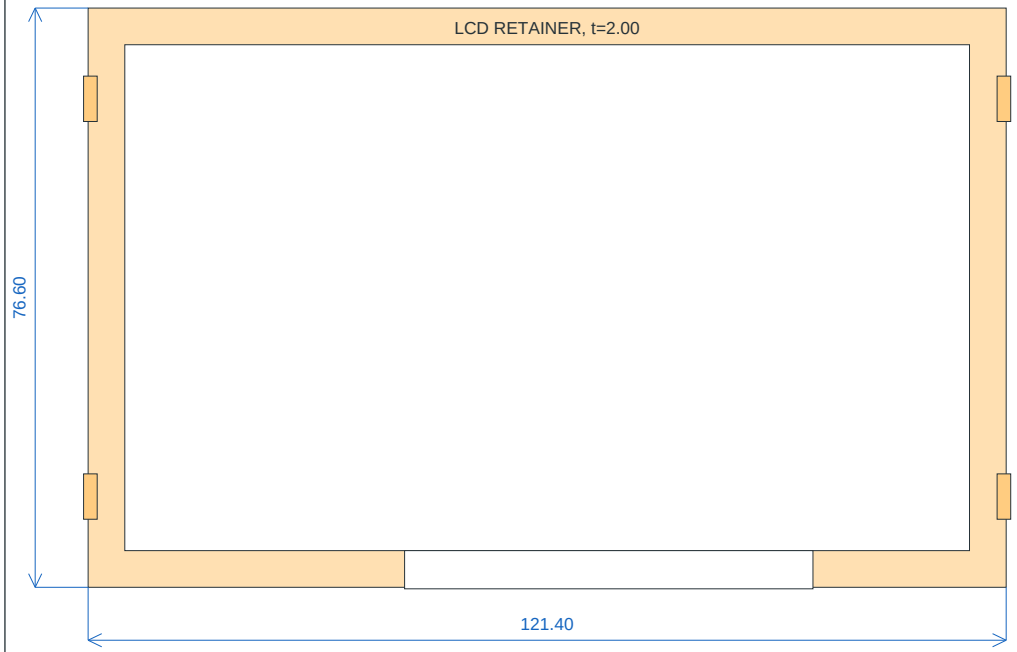
PRINT: A4 LANDSCAPE / ACTUAL SIZE (100%) / FIT TO PAGE OFF. VERIFY THE 100 mm LINE.

Printed parts B/C

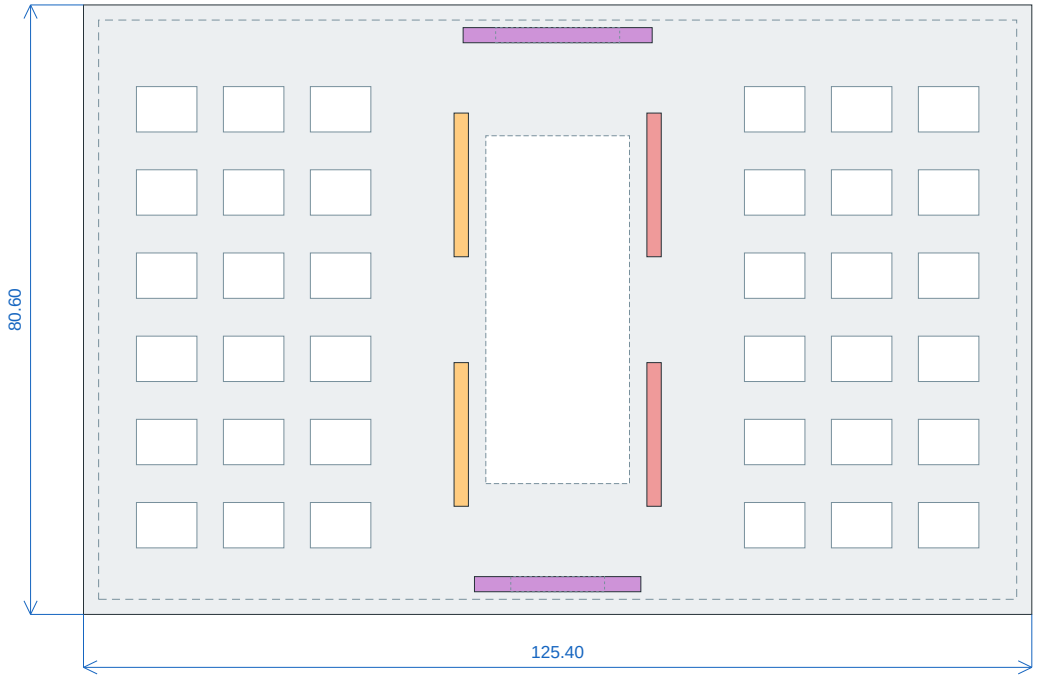
LCD retainer and rear cover with integrated PCB carrier — true 1:1

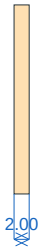
UNITS mm SCALE 1:1

B LCD RETAINER



C REAR COVER + PCB CARRIER





RETAINER THICKNESS

COVER PLATE THICKNESS

Orange: fixed guide. Red: two flexible PCB snap clips.

Centre aperture: microSD access, ventilation, and board removal.

Purple: connector bulkheads resist USB-C/HDMI insertion loads.

CALIBRATION 100.0 mm

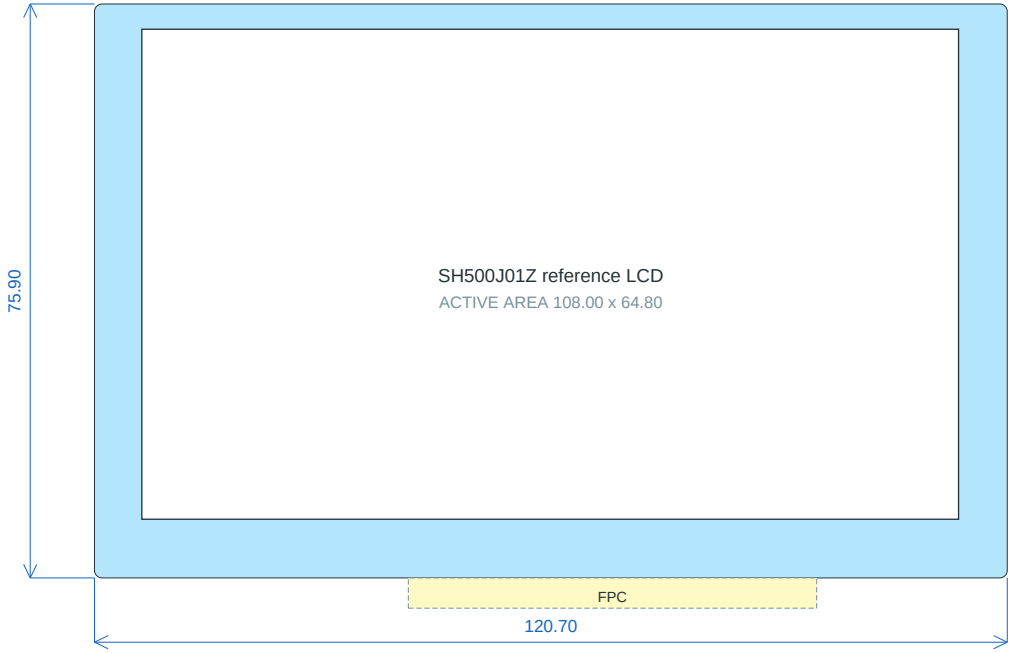
PRINT: A4 LANDSCAPE / ACTUAL SIZE (100%) / FIT TO PAGE OFF. VERIFY THE 100 mm LINE.

Physical hardware check templates

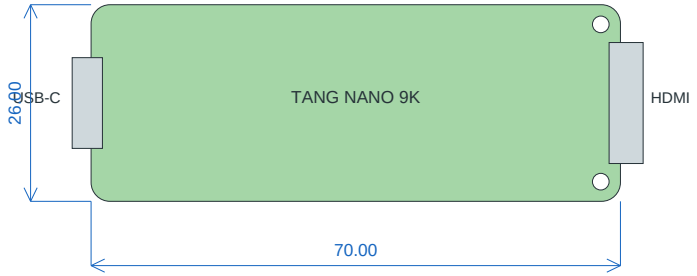
LCD, Tang Nano 9K, and panel cutout — all true 1:1

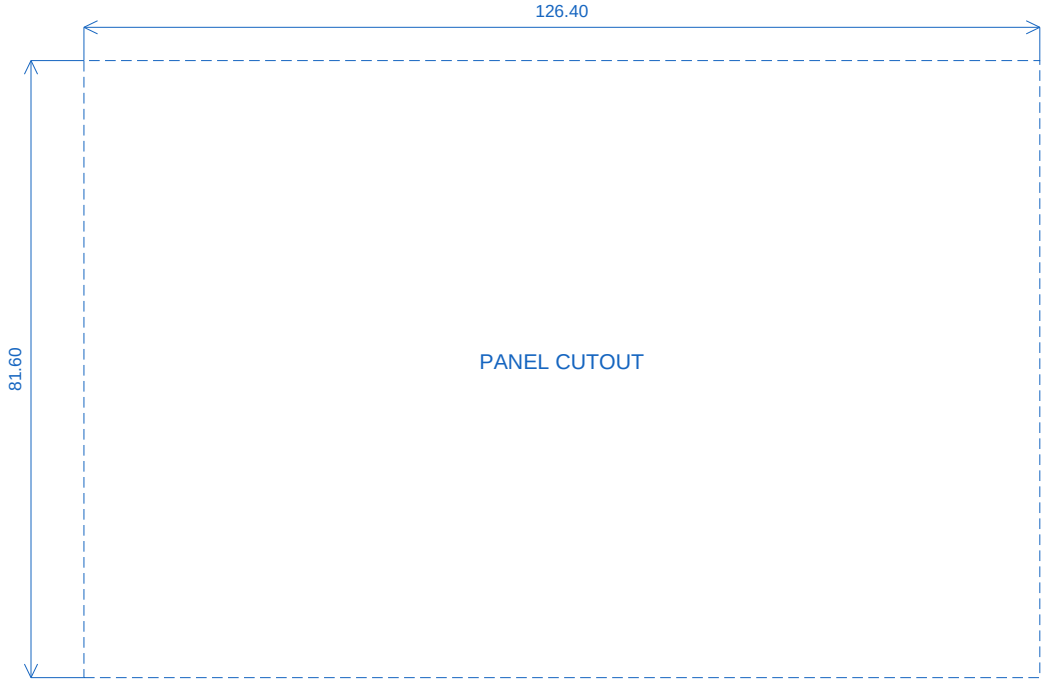
UNITS mm SCALE 1:1

LCD REFERENCE



TANG NANO 9K PCB





DIMENSION STATUS

VERIFIED: PCB outline 70.00 x 26.00 mm

VERIFIED: LCD outline 120.70 x 75.90 x 3.05 mm (SH500J01Z)

DESIGN VALUE: panel cutout 126.40 x 81.60 mm

CHECK HARDWARE: connector height/projection and LCD FPC shape

Do not print the STL if your LCD does not match this template.

CALIBRATION 100.0 mm

PRINT: A4 LANDSCAPE / ACTUAL SIZE (100%) / FIT TO PAGE OFF. VERIFY THE 100 mm LINE.

Assembly section index

Cutting planes are positioned to pass through the actual hooks, clips, connectors and service aperture

UNITS mm SCALE 1:1

REAR PROJECTION / SECTION LOCATIONS / SCALE 1:1

SECTION PURPOSE

A-A LONGITUDINAL Y-Z SECTION

USB-C / HDMI, PCB end stop, LCD, FPC route and rear cover

B-B PCB RETENTION X-Z SECTION

fixed lip, support shelf, PCB and flexible clip

C-C LCD HOOK X-Z SECTION

retainer cantilevers, stepped heads and chassis windows

D-D MICROSD SERVICE X-Z SECTION

PCB underside, service aperture and rear clearance

E-E HDMI-END M2 BOSS X-Z SECTION

two HDMI-end mounting bosses, pilot bores, PCB and deep rear shell

- FRONT CHASSIS
- LCD VERIFIED BODY
- LCD RETAINER
- PCB 70 x 26 x 1.6
- REAR COVER / CARRIER
- REFERENCE ENVELOPE — VERIFY ON HARDWARE

SOLID FILLS: evaluated directly from STL CSG geometry.

DASHED ORANGE: electronics envelope only; measure the physical board.

CALIBRATION 100.0 mm

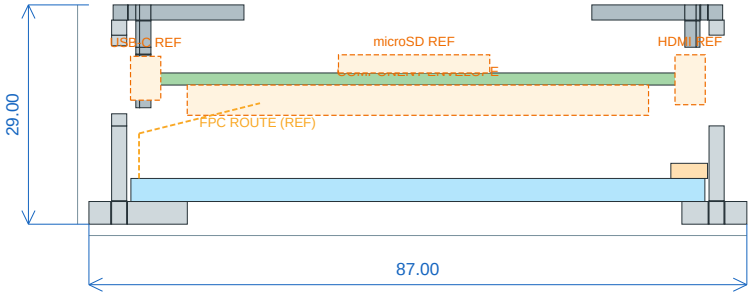
PRINT: A4 LANDSCAPE / ACTUAL SIZE (100%) / FIT TO PAGE OFF. VERIFY THE 100 mm LINE.

Exact assembly sections A-A / B-B

Solid geometry is evaluated from the STL source; all mechanical sections are true 1:1

UNITS mm SCALE 1:1

A-A Y-Z @ X=58.90 mm / USB-C → HDMI



WHAT THIS CUT PASSES THROUGH

- verified PCB edge-stop at both connector ends
- USB-C and HDMI wall openings
- LCD body, retainer plate and rear cover
- reference-only connector/component envelopes
- reference FPC routing corridor

FRONT CHASSIS

LCD VERIFIED BODY

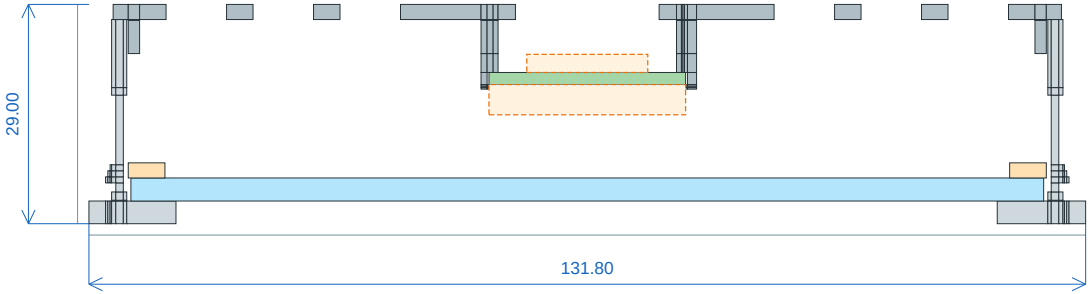
LCD RETAINER

PCB 70 x 26 x 1.6

REAR COVER / CARRIER

REFERENCE ENVELOPE — VERIFY ON HARDWARE

B-B X-Z @ Y=28.00 mm / FIXED LIP ← PCB → FLEX CLIP



The PCB is not floating:

z=18.40 support shelves under both edges

left fixed lip + right 1.20 mm cantilever clip

dashed volumes above/below PCB are unverified components

CALIBRATION 100.0 mm

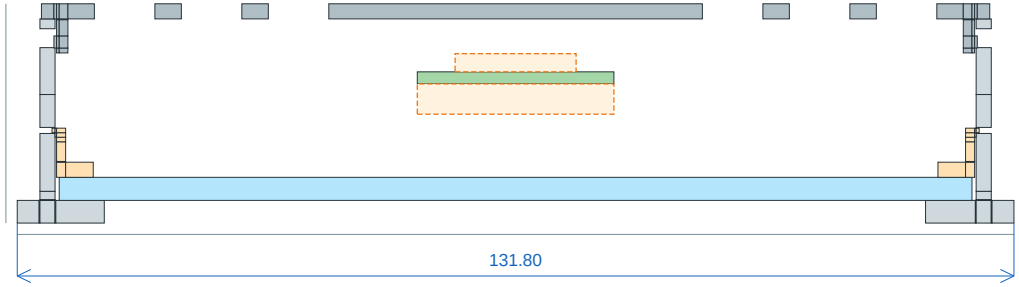
PRINT: A4 LANDSCAPE / ACTUAL SIZE (100%) / FIT TO PAGE OFF. VERIFY THE 100 mm LINE.

Exact assembly sections C-C / D-D

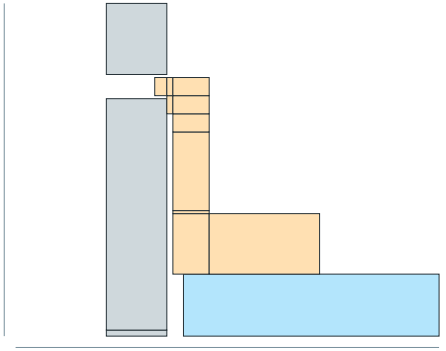
LCD hook engagement and microSD access are shown at their actual cutting planes

UNITS mm SCALE 1:1

C-C X-Z @ Y=17.20 mm / BOTH LCD HOOKS



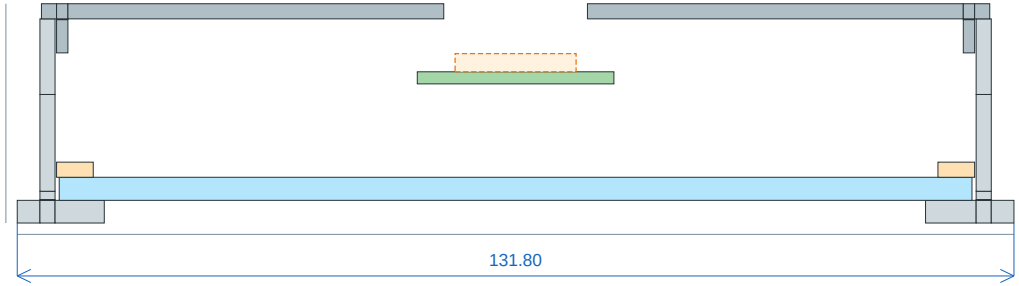
LEFT HOOK DETAIL 4:1



orange: retainer arm/head

grey gap: chassis window z=11.85..12.65

D-D X-Z @ Y=43.50 mm / microSD SERVICE APERTURE



At D-D the rear-cover plate is intentionally absent

under the PCB. This is the service aperture, not a hollow error.

The dashed microSD socket is a reference envelope;

verify its exact location and height on the physical board.

CALIBRATION 100.0 mm

PRINT: A4 LANDSCAPE / ACTUAL SIZE (100%) / FIT TO PAGE OFF. VERIFY THE 100 mm LINE.

Section interpretation and assembly

The section set distinguishes verified printable geometry from hardware values requiring measurement

UNITS mm SCALE 1:1

ASSEMBLY ORDER

1. Install the selected front chassis in the 126.40 x 81.60 panel cutout.
2. Place the LCD at z=3.00..6.05 with its display toward the bezel.
3. Push the retainer until all four C-C hook heads engage their wall windows.
4. Insert the PCB left edge below the B-B fixed lips and press the right edge into both clips.
5. Confirm the PCB rests on four shelves and between both connector bulkheads.
6. Connect the FPC, verify the measured connector/component envelopes, then latch the cover.

SECTION READING RULES

- Filled regions are exact intersections of the generated STL CSG solids.
- Empty regions may be intentional openings: display window, latch window, port opening or service aperture.
- Orange dashed regions are not printable geometry and are not dimensionally certified.
- A-A through E-E coordinates are measured from the bezel left/bottom/front datums.

MEASUREMENT HOLD POINTS

- ☐ USB-C shell: width, height, projection
- ☐ HDMI shell: width, height, projection
- ☐ microSD socket: x/y position and rear height
- ☐ maximum component height on both PCB faces
- ☐ LCD FPC exit point, width and bend radius
- ☐ cable overmould: USB-C 15.0 x 9.0, HDMI 23.0 x 12.0 max

Do not treat dashed envelopes as production dimensions.

- FRONT CHASSIS
- LCD VERIFIED BODY
- LCD RETAINER
- PCB 70 x 26 x 1.6
- REAR COVER / CARRIER
- REFERENCE ENVELOPE — VERIFY ON HARDWARE

CALIBRATION 100.0 mm

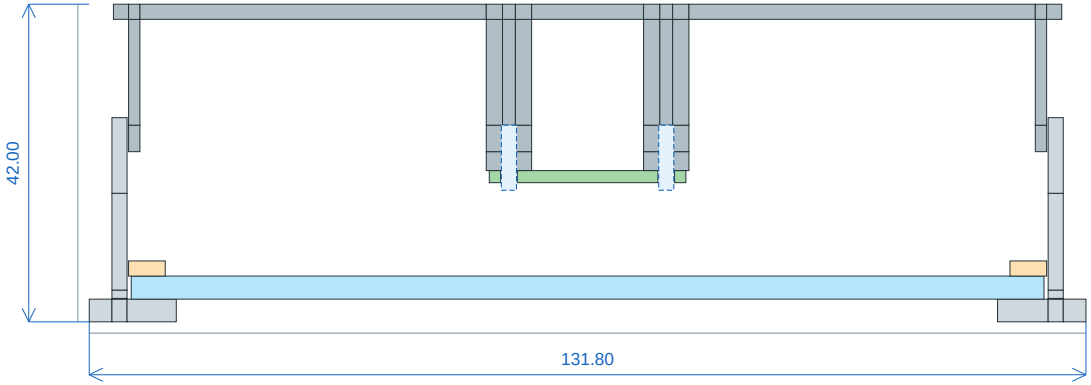
PRINT: A4 LANDSCAPE / ACTUAL SIZE (100%) / FIT TO PAGE OFF. VERIFY THE 100 mm LINE.

Expanded rear-cover sections

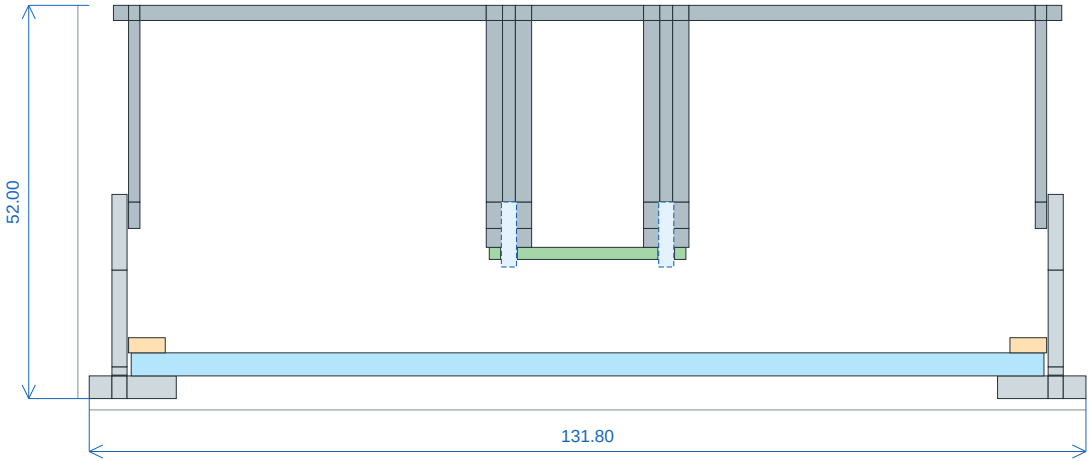
HDMI-end E-E cut through both M2 bosses; PCB and connector planes remain unchanged

UNITS mm SCALE 1:1

20 mm CLEARANCE VARIANT / E-E X-Z / SCALE 1:1



30 mm CLEARANCE VARIANT / E-E X-Z / SCALE 1:1



E-E CUT CONTENT

- two HDMI-end mounting holes only
- two 6.00 mm square printed bosses
- 1.70 x 1.70 mm M2 self-tapping pilot
- 6.00 mm effective thread depth
- optional M2x8 screws from PCB front side
- fixed lip and flex clips remain active

DEPTH DATUM

PCB rear surface: z=20.00 mm in every variant

20 mm cover inner face: z=40.00 mm

30 mm cover inner face: z=50.00 mm

Blue dashed shafts are screw references; grey bosses are STL geometry.

Verify the physical daughterboard outline and cable bend space.

CALIBRATION 100.0 mm

PRINT: A4 LANDSCAPE / ACTUAL SIZE (100%) / FIT TO PAGE OFF. VERIFY THE 100 mm LINE.

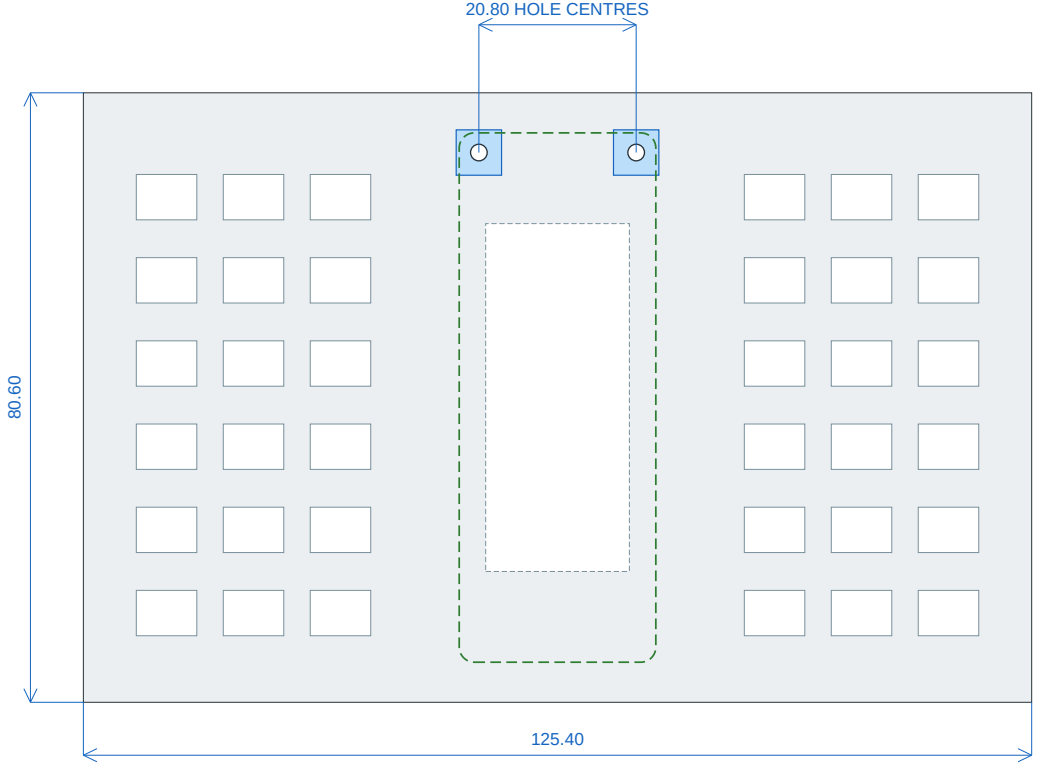
HDMI-end two-hole mounting layout

Rear projection at true 1:1; screw fixation is optional and supplements the snap carrier

SHEET 10/10

UNITS mm SCALE 1:1

REAR-COVER INBOARD VIEW / HDMI END AT TOP



Blue: printed bosses / Green dashed: Tang Nano 9K PCB

MOUNTING SPECIFICATION

PCB holes	2 only, HDMI end
Hole reference	diameter 2.20 mm
Hole-centre spacing	20.80 mm
Printed boss	6.00 x 6.00 mm
Pilot bore	1.70 x 1.70 mm square
Recommended screw	M2x8 self-tapping
Rear clearance variants	20.00 / 30.00 mm
Overall case depth	42.00 / 52.00 mm

ASSEMBLY

1. Insert PCB under the fixed lips and engage both flex clips.
2. Align the two HDMI-end PCB holes with the printed pilots.
3. Install M2x8 screws without compressing or bowing the PCB.
4. Confirm 20/30 mm daughterboard height and cable clearance.

Do not use screws longer than required; the pilot thread depth is 6.00 mm.

CALIBRATION 100.0 mm

PRINT: A4 LANDSCAPE / ACTUAL SIZE (100%) / FIT TO PAGE OFF. VERIFY THE 100 mm LINE.